

Assignment 5 - Hierarchical Clustering

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```
library(cluster)
library(factoextra)
```

```
## Loading required package: ggplot2
```

```
## Welcome to factoextra!
```

```
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practice
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

Import Dataset

```
cereals <- read.csv("Cereals.csv")
```

```
head(cereals)
```

```
##           name mfr type calories protein fat sodium fiber carbo
## 1      100%_Bran   N    C       70        4   1   130  10.0   5.0
## 2 100%_Natural_Bran Q    C      120        3   5    15   2.0   8.0
## 3        All-Bran   K    C       70        4   1   260   9.0   7.0
## 4 All-Bran_with_Extra_Fiber K    C       50        4   0   140  14.0   8.0
## 5        Almond_Delight R    C      110        2   2   200   1.0  14.0
## 6 Apple_Cinnamon_Cheerios G    C      110        2   2   180   1.5  10.5
##      sugars potass vitamins shelf weight cups   rating
```

```
## 1      6    280      25      3      1 0.33 68.40297
## 2      8    135       0      3      1 1.00 33.98368
## 3      5    320      25      3      1 0.33 59.42551
## 4      0    330      25      3      1 0.50 93.70491
## 5      8     NA      25      3      1 0.75 34.38484
## 6     10     70      25      1      1 0.75 29.50954
```

```
dim(cereals)
```

```
## [1] 77 16
```

```
str(cereals)
```

```
## 'data.frame': 77 obs. of 16 variables:
## $ name : chr "100%_Bran" "100%_Natural_Bran" "All-Bran" "All-Bran_with_Extra_Fiber" ...
## $ mfr : chr "N" "Q" "K" "K" ...
## $ type : chr "C" "C" "C" "C" ...
## $ calories: int 70 120 70 50 110 110 110 130 90 90 ...
## $ protein : int 4 3 4 4 2 2 2 3 2 3 ...
## $ fat : int 1 5 1 0 2 2 0 2 1 0 ...
## $ sodium : int 130 15 260 140 200 180 125 210 200 210 ...
## $ fiber : num 10 2 9 14 1 1.5 1 2 4 5 ...
## $ carbo : num 5 8 7 8 14 10.5 11 18 15 13 ...
## $ sugars : int 6 8 5 0 8 10 14 8 6 5 ...
## $ potass : int 280 135 320 330 NA 70 30 100 125 190 ...
## $ vitamins: int 25 0 25 25 25 25 25 25 25 25 ...
## $ shelf : int 3 3 3 3 3 1 2 3 1 3 ...
## $ weight : num 1 1 1 1 1 1 1 1.33 1 1 ...
## $ cups : num 0.33 1 0.33 0.5 0.75 0.75 1 0.75 0.67 0.67 ...
## $ rating : num 68.4 34 59.4 93.7 34.4 ...
```

```
names(cereals)
```

```
## [1] "name" "mfr" "type" "calories" "protein" "fat"
## [7] "sodium" "fiber" "carbo" "sugars" "potass" "vitamins"
## [13] "shelf" "weight" "cups" "rating"
```

Check Missing Values

```
colSums(is.na(cereals))
```

```
## name mfr type calories protein fat sodium fiber
## 0 0 0 0 0 0 0 0
## carbo sugars potass vitamins shelf weight cups rating
## 1 1 2 0 0 0 0 0
```

Select Numerical Variables

```
cereals_numeric <- cereals[, c(
  "calories",
  "protein",
  "fat",
  "sodium",
  "fiber",
  "carbo",
  "sugars",
  "potass",
  "vitamins",
  "shelf",
  "weight",
  "cups",
  "rating"
)]
```

```
head(cereals_numeric)
```

```
##      calories protein fat sodium fiber carbo sugars potass vitamins shelf weight
## 1         70      4  1   130  10.0  5.0      6   280         25      3      1
## 2        120      3  5    15   2.0  8.0      8   135          0      3      1
## 3         70      4  1   260   9.0  7.0      5   320         25      3      1
## 4         50      4  0   140  14.0  8.0      0   330         25      3      1
## 5        110      2  2   200   1.0 14.0      8    NA         25      3      1
## 6        110      2  2   180   1.5 10.5     10    70         25      1      1
##      cups  rating
## 1 0.33 68.40297
## 2 1.00 33.98368
## 3 0.33 59.42551
## 4 0.50 93.70491
## 5 0.75 34.38484
## 6 0.75 29.50954
```

```
summary(cereals_numeric)
```

```
##      calories      protein      fat      sodium
##  Min.   : 50.0    Min.   :1.000    Min.   :0.000    Min.   : 0.0
## 1st Qu.:100.0    1st Qu.:2.000    1st Qu.:0.000    1st Qu.:130.0
##  Median :110.0    Median :3.000    Median :1.000    Median :180.0
##  Mean   :106.9    Mean   :2.545    Mean   :1.013    Mean   :159.7
## 3rd Qu.:110.0    3rd Qu.:3.000    3rd Qu.:2.000    3rd Qu.:210.0
##  Max.   :160.0    Max.   :6.000    Max.   :5.000    Max.   :320.0
##
##      fiber      carbo      sugars      potass
##  Min.   : 0.000    Min.   : 5.0    Min.   : 0.000    Min.   : 15.00
## 1st Qu.: 1.000    1st Qu.:12.0    1st Qu.: 3.000    1st Qu.: 42.50
##  Median : 2.000    Median :14.5    Median : 7.000    Median : 90.00
##  Mean   : 2.152    Mean   :14.8    Mean   : 7.026    Mean   : 98.67
## 3rd Qu.: 3.000    3rd Qu.:17.0    3rd Qu.:11.000    3rd Qu.:120.00
##  Max.   :14.000    Max.   :23.0    Max.   :15.000    Max.   :330.00
##
##      vitamins      shelf      weight      cups
##  NAs      :1      NAs      :1      NAs      :2
```

```
## Min.    : 0.00    Min.    :1.000    Min.    :0.50    Min.    :0.250
## 1st Qu.: 25.00    1st Qu.:1.000    1st Qu.:1.00    1st Qu.:0.670
## Median : 25.00    Median :2.000    Median :1.00    Median :0.750
## Mean   : 28.25    Mean   :2.208    Mean   :1.03    Mean   :0.821
## 3rd Qu.: 25.00    3rd Qu.:3.000    3rd Qu.:1.00    3rd Qu.:1.000
## Max.   :100.00    Max.   :3.000    Max.   :1.50    Max.   :1.500
##
##      rating
## Min.    :18.04
## 1st Qu.:33.17
## Median :40.40
## Mean   :42.67
## 3rd Qu.:50.83
## Max.   :93.70
##
```

Replace Missing Values

```
cereals_numeric$carbo[is.na(cereals_numeric$carbo)] <-
  median(cereals_numeric$carbo, na.rm = TRUE)

cereals_numeric$sugars[is.na(cereals_numeric$sugars)] <-
  median(cereals_numeric$sugars, na.rm = TRUE)

cereals_numeric$potass[is.na(cereals_numeric$potass)] <-
  median(cereals_numeric$potass, na.rm = TRUE)

colSums(is.na(cereals_numeric))
```

```
## calories  protein      fat  sodium  fiber  carbo  sugars  potass
##          0         0        0        0      0        0         0
## vitamins  shelf  weight    cups  rating
##          0         0        0        0      0
```

Standardize Variables

```
cereals_scaled <- scale(cereals_numeric)

head(cereals_scaled)
```

```
##      calories  protein      fat  sodium  fiber  carbo
## [1,] -1.8929836  1.3286071 -0.01290349 -0.3539844  3.29284661 -2.5243405
## [2,]  0.6732089  0.4151897  3.96137277 -1.7257708 -0.06375361 -1.7514808
## [3,] -1.8929836  1.3286071 -0.01290349  1.1967306  2.87327158 -2.0091007
## [4,] -2.9194605  1.3286071 -1.00647256 -0.2346986  4.97114672 -1.7514808
## [5,]  0.1599704 -0.4982277  0.98066557  0.4810160 -0.48332864 -0.2057614
## [6,]  0.1599704 -0.4982277  0.98066557  0.2424445 -0.27354112 -1.1074310
##      sugars  potass  vitamins  shelf  weight  cups
```

```
## [1,] -0.2358694  2.6126578 -0.1453172  0.9515734 -0.1967771 -2.1100340
## [2,]  0.2239266  0.5260824 -1.2642598  0.9515734 -0.1967771  0.7690100
## [3,] -0.4657674  3.1882648 -0.1453172  0.9515734 -0.1967771 -2.1100340
## [4,] -1.6152574  3.3321665 -0.1453172  0.9515734 -0.1967771 -1.3795303
## [5,]  0.2239266 -0.1214755 -0.1453172  0.9515734 -0.1967771 -0.3052601
## [6,]  0.6837226 -0.4092790 -0.1453172 -1.4507595 -0.1967771 -0.3052601
##           rating
## [1,]  1.8321876
## [2,] -0.6180571
## [3,]  1.1930986
## [4,]  3.6333849
## [5,] -0.5894990
## [6,] -0.9365625
```

```
summary(cereals_scaled)
```

```
##      calories      protein      fat      sodium
##  Min.   :-2.9195  Min.   :-1.4116  Min.   :-1.0065  Min.   :-1.9047
## 1st Qu.: -0.3533  1st Qu.: -0.4982  1st Qu.: -1.0065  1st Qu.: -0.3540
## Median :  0.1600  Median :  0.4152  Median : -0.0129  Median :  0.2424
## Mean   :  0.0000  Mean   :  0.0000  Mean   :  0.0000  Mean   :  0.0000
## 3rd Qu.:  0.1600  3rd Qu.:  0.4152  3rd Qu.:  0.9807  3rd Qu.:  0.6003
## Max.   :  2.7262  Max.   :  3.1554  Max.   :  3.9614  Max.   :  1.9124
##      fiber      carbo      sugars      potass
##  Min.   :-0.90290  Min.   :-2.52434  Min.   :-1.615257  Min.   :-1.2007
## 1st Qu.: -0.48333  1st Qu.: -0.72100  1st Qu.: -0.925563  1st Qu.: -0.7690
## Median : -0.06375  Median : -0.07695  Median : -0.005971  Median : -0.1215
## Mean   :  0.00000  Mean   :  0.00000  Mean   :  0.000000  Mean   :  0.0000
## 3rd Qu.:  0.35582  3rd Qu.:  0.56710  3rd Qu.:  0.913621  3rd Qu.:  0.3102
## Max.   :  4.97115  Max.   :  2.11282  Max.   :  1.833213  Max.   :  3.3322
##      vitamins      shelf      weight      cups
##  Min.   :-1.2643  Min.   :-1.4508  Min.   :-3.5195  Min.   :-2.4538
## 1st Qu.: -0.1453  1st Qu.: -1.4508  1st Qu.: -0.1968  1st Qu.: -0.6490
## Median : -0.1453  Median : -0.2496  Median : -0.1968  Median : -0.3053
## Mean   :  0.0000  Mean   :  0.0000  Mean   :  0.0000  Mean   :  0.0000
## 3rd Qu.: -0.1453  3rd Qu.:  0.9516  3rd Qu.: -0.1968  3rd Qu.:  0.7690
## Max.   :  3.2115  Max.   :  0.9516  Max.   :  3.1260  Max.   :  2.9176
##      rating
##  Min.   :-1.7529
## 1st Qu.: -0.6757
## Median : -0.1613
## Mean   :  0.0000
## 3rd Qu.:  0.5811
## Max.   :  3.6334
```

```
dim(cereals_scaled)
```

Create Distance Matrix

```
distance_matrix <- dist(
  cereals_scaled,
  method = "euclidean"
)
```

```
distance_matrix
```

##	1	2	3	4	5	6	7
## 2	7.5544859						
## 3	1.9075753	7.7994968					
## 4	3.4157603	10.0595394	4.0482021				
## 5	6.7587226	4.4993106	6.4370095	9.1497968			
## 6	7.0918084	4.8662196	6.9276794	9.5922655	2.6645515		
## 7	7.5853473	5.9292664	7.6407712	9.8399263	3.2555939	2.8777831	
## 8	7.3939571	5.3583461	7.0300360	9.5341024	2.8294734	4.1302412	4.5488612
## 9	5.7295413	6.1849273	5.4701695	7.5426335	3.3386260	2.8427383	3.9346185
## 10	4.1776812	6.4778812	3.7762778	5.9871765	3.6351645	4.6476142	4.5923443
## 11	8.3051471	5.2934729	8.0010101	10.8665615	2.4216589	2.1083290	3.0172801
## 12	7.9777854	6.7211739	7.4918552	9.3131335	5.4459857	5.4184552	6.1643479
## 13	8.3292721	4.6535174	8.0125652	10.8407390	2.3089105	2.2637152	3.8570923
## 14	5.6979648	4.3984832	5.5903494	8.2914355	1.7395017	3.1267368	3.9159721
## 15	8.1430278	5.5693868	7.9391989	10.4856242	2.6794429	2.3913480	1.7968672
## 16	8.9559592	7.7855151	8.4687151	10.3483664	4.3193832	4.4079489	4.4514032
## 17	8.4556335	7.7738262	7.9615280	9.7113640	4.3288340	4.4091508	4.5441062
## 18	7.9038466	6.1001097	8.0168698	9.9664559	3.3223826	3.1495736	1.2412564
## 19	8.0966466	5.5449788	7.8791542	10.4439072	2.6638106	2.3876875	1.8410664
## 20	4.9535019	3.6969482	4.8904246	7.7472056	2.7180821	3.5847931	4.8804037
## 21	7.4990309	6.9065964	7.5145882	8.5117880	4.7101333	5.4096296	5.0361879
## 22	8.0617896	6.9893149	7.7299938	9.4620931	3.3573402	4.7870360	4.0881735
## 23	5.8348531	4.8157991	5.7485596	8.3187036	1.7176263	2.8377791	2.5737438
## 24	6.8711651	6.4644201	6.5596707	8.6277660	2.5071802	4.0286536	3.4846916
## 25	7.4904844	5.0697993	7.5337638	9.8479437	2.6594292	2.2178731	1.0221382
## 26	7.9864650	6.8488629	7.7887086	10.0934781	3.4567180	2.4875643	2.3076074
## 27	5.7788737	5.9226694	6.2579749	7.5421583	3.9892569	4.1509754	3.4894747
## 28	5.2040638	4.6771306	5.0239090	7.6986333	3.1967541	4.0335066	4.6315944
## 29	5.8506928	6.7565879	5.4114161	7.9306233	4.0196516	4.8053335	4.5255654
## 30	7.8682266	5.5846249	7.8040284	10.2702645	2.4562942	2.2372413	1.9447938
## 31	7.8007480	6.3049965	8.0269721	10.0681786	4.1950681	3.0339687	1.7760252
## 32	8.1152862	6.2807971	7.6107717	10.3433041	2.3360052	2.6027900	3.0732549
## 33	5.7745613	5.1643197	5.7562392	7.6262335	2.3972638	3.7529655	3.5085599
## 34	5.8018406	7.0140786	5.6243027	7.6761993	3.7385749	4.8931489	5.0252476
## 35	5.8463327	4.2006482	5.9983101	8.3753019	3.1407070	4.1144722	5.3055426
## 36	8.1979952	5.0030965	7.9168739	10.5599642	2.3886300	2.1432568	2.7507633
## 37	6.8580607	5.7986477	6.4566005	9.2392713	2.9452123	1.6371654	2.8837099
## 38	9.0556745	6.8532654	8.8804659	10.8815247	4.2734334	3.5374076	2.4641692
## 39	8.1440851	6.8778228	7.9293458	9.9255078	3.8152153	4.8123489	4.4723022
## 40	8.4497984	7.4431762	8.1997391	10.3396087	4.7126074	5.6287015	5.4884786
## 41	9.3799576	6.9278663	8.9470604	10.6961725	4.3261236	4.9695498	4.6488678
## 42	5.5943278	4.5086955	5.5213739	8.0084659	2.6043102	2.7796170	3.8597578
## 43	7.7253247	5.2597184	7.5273304	10.0826843	2.3405961	2.1039230	1.4869454
## 44	6.9739238	5.4056259	7.2220039	8.6449075	4.0947663	4.4913684	4.2775975
## 45	7.3964913	3.8104852	7.3323154	9.6742982	3.7511247	4.6167751	4.9984766
## 46	7.4556011	4.0189707	7.2341747	9.7510014	3.5804789	4.4976047	4.9870615

## 47	8.2156864	6.0882699	8.0005145	10.4977722	4.7598200	5.4620828	5.7130653
## 48	7.0963093	5.8956424	6.7652218	8.8994052	2.9753095	2.3489969	3.0963021
## 49	7.6262742	5.6261856	7.3617498	9.9815915	1.9338010	2.1922460	2.5487425
## 50	7.5528208	5.9723221	7.1211930	9.4871290	3.5681259	4.8682181	5.3841387
## 51	6.3201322	6.6477242	6.1737347	7.4665957	3.7444154	4.9972506	4.4138294
## 52	6.7665609	4.8546810	6.5050922	9.4200869	2.5430536	3.5647121	4.2383299
## 53	5.3353317	6.1126670	4.9909722	7.7225348	4.4675082	5.0094842	5.1588888
## 54	8.5823575	8.4864917	8.0152352	9.9160356	4.8892250	5.9591017	5.6025598
## 55	8.4338118	8.0222920	8.7745553	9.4561893	6.4589963	7.0544152	6.4014731
## 56	7.5688860	7.6865205	7.9953141	8.6059857	6.4418325	6.9551881	6.3608283
## 57	5.2291076	5.4361889	5.1661222	7.5589835	2.8236640	3.9828233	4.1827168
## 58	6.2232164	5.0596614	6.5884224	8.3021705	4.7578100	4.2636229	5.0464270
## 59	5.8442269	6.0915601	5.4156772	7.9869687	4.0979646	4.2558621	4.6927155
## 60	5.0094326	4.3196765	4.9175023	7.7911083	2.1527189	3.1905199	4.0217640
## 61	5.7506768	6.2998065	6.1379281	7.6374120	3.8221209	4.7415467	4.2154695
## 62	9.4047791	7.9351929	9.0020007	10.6465488	4.6842386	4.8199732	4.7126993
## 63	8.9364793	7.8028716	8.4115670	10.3310952	4.3117424	4.4113589	4.4944535
## 64	7.0162524	7.1337958	7.3963896	7.8539575	5.6239849	5.4800355	5.3284689
## 65	6.3191369	7.4360409	6.6923824	7.0978490	5.8273312	5.8530327	6.0130901
## 66	6.7304180	7.4607505	7.0510086	7.5894239	5.7281268	5.7882656	5.9069125
## 67	7.1621979	5.0903287	7.3803677	9.7982780	3.1067394	2.4223414	1.7116809
## 68	7.6454585	7.4148995	7.3704838	9.1015139	5.2889291	5.1306516	5.1539447
## 69	6.3401073	6.1779368	6.7223789	7.7209382	4.1082135	4.3742580	3.6116530
## 70	8.9294740	7.5648540	8.6062675	10.4726434	4.3457929	5.4986772	5.2796381
## 71	8.3521328	7.7559386	8.0287965	10.2572208	5.9541427	6.4835277	6.2913467
## 72	6.9980275	6.9598481	6.7199218	8.4892416	4.2005403	5.2106300	5.1576940
## 73	7.8930696	6.4572599	7.3751738	9.6479277	2.5281702	4.2715342	4.4022839
## 74	8.1892141	5.4977372	8.1146366	10.4459896	2.6676662	2.4704868	1.6311182
## 75	6.2117905	6.3782220	5.7963612	7.8970560	3.4723009	3.2942268	4.4148391
## 76	6.6361023	6.1086478	6.3565897	8.0913122	3.6354300	3.4479592	4.1062349
## 77	7.4203122	5.9626199	7.1376373	9.5124991	2.6882510	1.8851685	2.8598469
##	8	9	10	11	12	13	14
## 2							
## 3							
## 4							
## 5							
## 6							
## 7							
## 8							
## 9	4.3739183						
## 10	4.4530715	3.0108692					
## 11	4.0632190	4.2069731	5.4377141				
## 12	5.3184858	5.0141597	5.5159582	6.5633421			
## 13	3.9223191	4.2381291	5.6069719	1.2570727	6.3106593		
## 14	3.0803917	3.3440886	3.2202906	3.5704237	5.5115570	3.4188796	
## 15	4.3339527	3.9930815	5.0321066	1.6969504	6.3747293	2.5436144	3.9221578
## 16	4.7559608	3.7250550	5.1678917	4.6764931	4.7756234	4.7965509	5.1201404
## 17	4.8728531	3.3112454	4.7525465	4.8877682	4.5981205	4.9749882	5.0543338
## 18	4.6700214	3.8713826	4.7793437	3.1072104	6.5850075	3.7977569	4.0990666
## 19	4.3205724	3.9707667	4.9889510	1.7226705	6.3677279	2.5539473	3.9061998
## 20	3.8092631	3.8730583	3.6302931	4.2753534	5.9059357	3.9261491	1.7073619
## 21	5.0388736	4.0216388	4.2974534	6.1868631	4.9135686	6.1107190	4.6954277
## 22	3.9263367	4.0277845	4.0863957	4.6583546	5.2545724	4.7624032	4.1179853
## 23	3.6399013	3.1829265	2.9282382	3.0615489	5.9986995	3.3939940	2.0290745

## 24	3.6088930	3.1502907	2.9433478	4.0600622	5.5496892	4.2720728	2.9682693
## 25	4.1463003	3.7231963	4.6238866	2.4166897	5.8218017	3.0560656	3.4295380
## 26	4.6874297	3.1017822	4.8107858	2.6713210	6.3129319	3.4387626	4.1885721
## 27	4.7417511	3.2229437	3.3287936	5.2488328	5.6428854	5.4410894	3.4619624
## 28	2.7114599	4.0095064	3.4091913	4.5904804	5.8995892	4.5553877	2.7587729
## 29	3.1157297	4.2840856	3.2757933	5.1018974	6.2788355	5.5039250	3.9304184
## 30	4.1492422	3.6552457	4.8588918	1.6943334	6.5901443	2.4314532	3.3244129
## 31	5.4062864	4.0142069	5.2745980	3.7413046	6.6466022	4.4475913	4.4866034
## 32	3.8981968	3.4964639	4.6813982	1.7391790	6.0580648	2.2924056	3.6100724
## 33	3.4085579	2.9664825	2.4540874	4.3821147	4.6517618	4.3692650	2.3116327
## 34	4.1942702	3.6461035	2.9303226	5.3329407	6.1815256	5.4262107	2.8659406
## 35	3.8705787	4.2568344	4.3246625	4.6724969	6.2112796	4.1934932	1.7862565
## 36	3.9817578	4.0365745	5.2224832	1.2157048	6.0425658	1.6218934	3.7608282
## 37	4.0136838	2.6505116	4.0450947	2.8756937	4.6566823	3.2982973	3.2849232
## 38	5.3596050	4.3544826	5.7217493	3.6045354	6.0585640	4.1990583	5.4152543
## 39	4.5957641	4.8745174	5.0241048	4.6801735	6.1256461	4.7236472	4.4294878
## 40	3.6209709	5.7023076	5.6700815	5.5057824	6.6016866	5.5795693	4.8420427
## 41	4.8231165	4.8375121	5.6495192	4.9637143	4.5372647	4.8841465	5.5188122
## 42	3.6113334	2.8190741	3.3010866	4.0624197	4.1466944	3.9039233	1.8306840
## 43	3.9122786	3.6653989	4.5681772	1.9830079	5.5549790	2.6708381	3.4152849
## 44	4.6666416	4.0567893	4.3478999	5.5267691	4.4987454	5.4014677	3.7628391
## 45	3.5570095	5.3694072	5.2638097	4.8997710	5.3121401	4.6374683	3.6726516
## 46	3.3653086	5.2940575	5.1841173	4.6847824	5.1045691	4.4335264	3.6550119
## 47	2.6107253	6.0806226	6.0353147	5.3052809	7.1020995	5.3330396	4.7388486
## 48	4.1010016	1.9237443	3.8391047	3.4306721	4.3197616	3.5209645	3.7078796
## 49	3.2967413	3.2378457	4.3169204	1.9650245	5.5772254	2.3970448	2.6512650
## 50	1.2149281	4.6933608	4.6982535	4.8488634	5.5815175	4.6378072	3.6301309
## 51	4.3638608	3.4878938	2.7113303	5.5914884	4.7145393	5.6307368	3.8504938
## 52	1.8885188	4.3698977	4.3408476	3.5137896	6.0667564	3.5178602	2.2419741
## 53	3.8286901	4.8324991	3.8900033	5.4622978	6.8311026	5.7281915	4.2173508
## 54	5.2690767	5.3406742	5.1580084	5.9281085	5.6751791	6.0477435	5.4527286
## 55	8.5640249	6.3229177	6.3620345	7.4591106	8.3352733	7.4776308	6.5983402
## 56	8.5139720	6.1851841	5.9074289	7.6033600	7.9266853	7.6354678	6.3673486
## 57	3.6108892	3.3652143	2.5536451	4.7149208	5.1084189	4.7430377	1.6228422
## 58	5.0862931	4.0164142	4.8473015	5.8081338	4.7892598	5.6452450	3.8390600
## 59	3.2045316	3.8686102	3.7286944	4.9797833	5.7931366	5.1794751	4.0133173
## 60	3.6298630	3.3688957	2.9816439	3.8281138	5.7830154	3.7565741	1.0154841
## 61	4.8745973	3.7322964	3.3002531	5.2965606	6.9407336	5.4471878	3.1747723
## 62	5.2076989	4.1286525	5.6268072	4.9605994	5.4890010	5.0150241	5.6257390
## 63	4.7441691	3.7144698	5.1350080	4.6720385	4.7507145	4.7925223	5.1295676
## 64	6.7544887	3.9578648	4.8273458	6.6917667	6.0975920	6.6476945	5.4926256
## 65	6.1624040	3.8773335	4.4689434	7.1733371	6.0055166	7.0831998	5.2123363
## 66	6.0644696	3.9179564	4.6384386	7.0350143	5.9393139	6.9416596	5.1704017
## 67	4.5797770	4.0487279	4.8497084	2.7243300	6.6988071	3.4118234	3.3222251
## 68	5.2723345	4.5554003	4.8786422	6.2867145	2.5719477	6.4014882	5.0946806
## 69	5.0606471	3.2132634	3.5950651	5.3233802	5.7808100	5.4490665	4.0793426
## 70	4.8616124	5.2913633	5.5736232	5.2778553	6.1151806	5.1893532	5.0042652
## 71	4.6980277	6.7324656	6.3324843	6.5173423	7.5599534	6.7218290	6.1758000
## 72	4.7808396	4.5764018	4.1960192	5.6162501	5.3525816	5.5522821	4.4124876
## 73	3.2996982	3.7899185	4.0747535	3.9611230	5.2950193	3.8468259	3.3049176
## 74	4.2769583	3.9016883	5.0466357	1.9698324	6.3098345	2.6344991	3.8137753
## 75	4.0678725	1.4717213	3.2068698	4.5727689	3.9870317	4.4800950	3.3554208
## 76	4.2148078	2.0256665	3.5163392	4.7355546	3.3856002	4.6356525	3.8566586
## 77	3.7744514	2.1809412	4.2017225	2.7324170	4.9430469	2.9552217	3.2549181

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## 16	4.3092305						
## 17	4.4710011	0.8321926					
## 18	1.8844725	4.1730614	4.2809589				
## 19	0.1459223	4.3335281	4.4915093	1.9396994			
## 20	4.7360872	6.3161400	6.1556263	5.1292072	4.6987210		
## 21	5.5546388	3.7119786	3.5225798	4.6828609	5.5564701	5.7559083	
## 22	4.1733007	2.5848787	2.6067745	3.7225065	4.1981553	5.4237103	3.0625543
## 23	2.6379459	4.9293827	4.8250513	2.7860353	2.5994532	2.8054541	4.7478863
## 24	3.6244322	3.2340666	3.1388520	3.2376363	3.6233499	4.3516854	3.2268775
## 25	1.4460931	4.4472477	4.5331554	1.5490728	1.4995495	4.2745000	5.0234720
## 26	2.2053993	3.3485722	3.4659837	2.1304295	2.2448383	5.2284808	5.0569089
## 27	4.5314946	4.8707877	4.6823461	3.3895191	4.5268149	4.3183424	3.1083917
## 28	4.6652208	6.1325785	6.0100261	4.9394204	4.6095221	2.4188790	5.6735251
## 29	4.8528148	5.6995756	5.5931460	4.9062858	4.8036418	4.3198619	5.7240468
## 30	1.3746148	4.2402272	4.4093518	1.7074920	1.4329443	4.3792557	5.2046389
## 31	2.7395454	4.9264218	5.0127522	2.0238649	2.7625437	5.3933735	5.2475642
## 32	2.0115693	3.3515737	3.5125259	3.0086091	2.0280035	4.6369933	5.3252539
## 33	3.9560586	4.1565706	3.9169638	3.5259275	3.9558231	3.2825532	3.1090943
## 34	5.4503735	4.8531468	4.6941893	4.9705954	5.4487116	4.0374956	4.1677972
## 35	5.2819989	6.1884350	6.1194895	5.3076209	5.2738922	1.8810885	5.3404092
## 36	1.3748510	4.3842212	4.5355742	2.8011164	1.3964037	4.3290539	5.8133261
## 37	2.8505399	3.8505541	3.8233602	3.4779147	2.8338378	3.9725567	5.1190794
## 38	2.2766918	3.5688429	3.7449908	2.2801830	2.3039406	6.2925009	5.1643903
## 39	4.3535266	4.7366365	4.7768288	4.3291998	4.3590459	5.2757702	5.6426733
## 40	5.5359606	5.6652085	5.8547469	5.5241921	5.5271886	5.5992854	6.4403361
## 41	4.2952770	2.6908544	2.8096252	4.4043254	4.3115759	6.4359859	4.1200445
## 42	4.1364536	4.7239332	4.5467578	4.2445088	4.1284036	2.4685299	4.2511373
## 43	0.9839665	4.0067214	4.1460166	1.9710534	1.0016278	4.3254368	5.0415242
## 44	4.9282563	4.6369195	4.5581691	4.2395750	4.9248339	4.6767557	2.6029692
## 45	4.9789925	6.2355242	6.3869766	5.2988151	4.9381168	3.6661358	5.7374948
## 46	4.8475205	6.0492117	6.2034156	5.3470104	4.8044521	3.6486668	5.8704189
## 47	5.5990982	6.7473547	6.9641900	5.9082400	5.5643901	5.0012577	6.8936752
## 48	2.9055306	2.6605849	2.4268779	3.0636681	2.8945279	4.4470275	3.8467227
## 49	2.2077548	3.4250173	3.6558336	2.5870102	2.2321207	3.9508442	4.6145508
## 50	5.1814600	5.0039715	5.1299300	5.4060454	5.1594641	4.2432088	5.1935700
## 51	4.9356263	3.8253339	3.4362872	4.2770178	4.9359805	4.8279581	2.3968584
## 52	4.0508287	5.5649316	5.6753988	4.5339221	4.0234171	2.8558612	5.7165631
## 53	5.3137544	6.9175124	6.7880276	5.6035446	5.2402579	3.9411900	6.6781279
## 54	5.6216019	4.3461788	4.2706140	5.6273824	5.6329190	6.4384662	5.7723962

## 55	6.7432345	6.8774220	6.5796832	5.9433783	6.7667688	7.3382754	5.3830910
## 56	6.8736484	7.1691343	6.8112992	6.1119210	6.8867835	6.9396087	5.4025583
## 57	4.6838663	5.0219847	4.8617352	4.4591923	4.6716554	2.7933656	4.0149968
## 58	5.6694187	5.8261588	5.7106137	5.3125872	5.6597353	4.1998610	4.3113454
## 59	4.7835506	5.7196566	5.6097717	5.0563999	4.7110924	4.0258310	5.7651654
## 60	4.0497603	5.6723477	5.5193502	4.3221020	4.0160028	1.3423047	5.1487866
## 61	4.8421918	5.4315907	5.2581103	3.8804763	4.8324568	4.1973986	3.7814411
## 62	4.4373209	1.2247476	1.5125808	4.1558496	4.4588132	6.7535745	3.7712321
## 63	4.3128784	0.1972428	0.8373776	4.2383933	4.3320265	6.3057831	3.7848018
## 64	5.9308205	5.0155814	4.6266083	4.8727067	5.9318653	6.1430671	2.8205978
## 65	6.6899733	5.3350592	4.9819252	5.6922444	6.6785671	5.8480028	2.7378609
## 66	6.5601499	5.0481397	4.7416165	5.5526473	6.5543996	5.9432748	2.4353732
## 67	2.2598811	5.4256373	5.5058012	2.1812661	2.2846757	4.0648590	5.6650437
## 68	5.9033095	4.1411099	4.0233748	5.6344758	5.9139732	5.9875728	4.0885469
## 69	4.4006568	4.4764689	4.1935861	3.1749859	4.4009448	4.8917712	2.7388230
## 70	5.0656135	4.3656397	4.4633549	5.0052944	5.0813430	6.0101070	5.4984559
## 71	6.2814475	7.4450354	7.5172802	6.5822032	6.2268196	6.2909873	8.0235596
## 72	5.2751227	5.0638537	4.8577573	5.1846021	5.2633497	4.9732252	5.3900456
## 73	4.0190427	2.8962493	2.9988709	4.1416191	4.0261259	4.6294280	3.7630761
## 74	0.8128994	4.0816835	4.2568732	1.3604626	0.9075406	4.7614258	5.1105011
## 75	4.5158420	3.1185075	2.7507873	4.4430194	4.5009811	4.0993349	3.5167540
## 76	4.2983071	2.8735187	2.4746947	4.0918877	4.2858339	4.5312821	2.9840153
## 77	2.6399736	2.7078153	2.7790261	2.8133972	2.6488470	4.2906687	4.1572778
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## 23	3.8700905						
## 24	1.7044899	2.6001214					
## 25	4.0901808	2.3024224	3.5136561				
## 26	3.9582210	3.3432508	3.3640512	2.4567607			
## 27	4.0800604	3.1498963	3.2427779	3.5676090	4.1431322		
## 28	5.2228069	2.9833558	4.2908486	4.2662327	5.1341018	4.2679556	
## 29	4.8786359	3.5710081	4.0250445	4.6250309	4.8476213	4.5867856	2.3747981
## 30	3.9950943	2.4352987	3.2182138	1.6095915	1.8399346	4.0028524	4.5673637
## 31	5.0783437	3.4219488	4.3054973	2.0796870	2.5627628	3.4198998	5.3213506

## 32	3.4853186	3.0147790	2.9475841	2.8098012	2.0426636	4.9319910	4.7400923
## 33	2.7071699	2.2892371	2.0248525	3.2579469	4.0940516	2.4208567	3.4153787
## 34	3.7816130	3.6971628	2.6809898	5.0331861	4.6744117	3.6045959	4.2998364
## 35	5.2238045	3.5486132	4.2920600	4.7262257	5.4138272	4.1950538	3.4315320
## 36	4.2974154	2.9798948	3.9577250	2.0578887	2.7708882	5.0148495	4.4436759
## 37	4.4173185	2.9949389	3.6673337	2.6349147	2.4632230	4.0877227	4.0277357
## 38	4.1629200	4.1744348	4.2292085	2.5993008	2.5499075	4.7525228	5.9404815
## 39	3.8575006	4.1195698	3.7736984	4.2730619	4.7377855	4.8729482	5.2551447
## 40	4.9513332	5.0969133	4.7940400	5.3570118	5.6353670	5.5903961	4.6711790
## 41	2.7870326	5.0443420	3.9018586	4.4248214	4.6183396	5.3878278	6.1736057
## 42	4.3145331	2.7139656	3.4270194	3.3566637	4.1705353	3.1072331	3.3040712
## 43	3.8314377	2.3230490	3.2734277	1.0034537	2.2952207	4.0110000	4.2826755
## 44	3.9935530	3.9462686	3.6494440	4.0518919	4.9664175	2.3070911	4.9042183
## 45	5.3802566	4.2099940	5.1046520	4.4528821	5.8813586	4.9168171	3.4120587
## 46	5.2690659	4.1792916	5.0264317	4.4367571	5.7662548	5.1751549	3.3489621
## 47	6.0878339	5.1013828	5.6680500	5.4407199	6.0374305	6.0544586	3.2755512
## 48	3.3229303	3.1635566	2.9963102	2.8070892	2.4729483	3.5904926	4.4120250
## 49	3.2833889	2.5124980	2.5698145	2.2374382	2.0425784	3.8762122	4.1373164
## 50	4.2177701	4.3916892	4.0188263	5.0416965	5.2850283	5.1606278	3.0690619
## 51	2.2731020	3.6269560	2.1673587	4.4178584	4.7326950	2.9949084	4.6758300
## 52	4.7941293	2.9764887	3.8912579	3.8223013	4.4878132	4.6055256	2.2355218
## 53	6.1453280	3.8810923	5.1646811	5.0724998	5.6924910	5.1478199	1.8382035
## 54	3.7482331	5.3201857	4.0536973	5.5993099	5.4965084	5.8329389	6.2701803
## 55	5.9511676	6.0201684	5.6037462	6.3885784	6.7385140	5.4920904	8.4128014
## 56	6.2102256	5.8308306	5.7176535	6.3495865	6.9183989	5.1843556	8.0583464
## 57	3.9364871	2.6191331	2.7426590	3.9849622	4.5755157	2.8573464	3.3173982
## 58	5.7623083	4.5727573	5.0664898	4.6899332	5.4218286	3.2607380	4.6705425
## 59	5.4113622	3.7763186	4.5917498	4.5760446	4.8586182	4.5936545	2.0021649
## 60	4.7302884	1.8421351	3.4280905	3.5637563	4.4448537	3.5839327	2.5726177
## 61	4.1573797	2.9969323	2.8601055	4.2645483	4.5231424	2.1364303	4.4551244
## 62	2.7550661	5.2381559	3.5703192	4.6981141	3.6465829	5.0892656	6.5347813
## 63	2.6304806	4.9290398	3.2484322	4.4893725	3.3748689	4.9397632	6.1040493
## 64	4.8558206	5.1712739	4.5691902	5.3198618	5.2746147	3.1549874	6.5254370
## 65	5.1472483	5.4559758	4.6580792	6.0105490	5.7633864	3.0569307	5.9085250
## 66	4.9125944	5.4452226	4.4839760	5.9029392	5.6108752	3.0645256	6.0382046
## 67	5.0217338	2.3697163	4.0748753	1.4446180	2.8671738	3.6118847	4.2509660
## 68	4.5956947	5.4253585	4.6652370	5.1614579	5.3254268	4.4557121	5.8769721
## 69	3.6488869	3.3972298	3.1121392	3.6597824	4.1055593	1.4738064	4.8179062
## 70	3.5808508	4.9659714	3.9043661	5.0830817	5.2080935	5.4508074	6.0314300
## 71	6.8541239	5.8546236	6.4712086	6.2134187	6.8420492	6.7562417	4.5588727
## 72	4.1351040	4.3326152	3.9563848	4.9640250	5.4593245	4.6853427	4.9505641
## 73	1.7144096	3.5643925	1.7230134	4.1648354	3.9129184	4.5117881	4.7495137
## 74	3.8395076	2.6653448	3.3884594	1.2109663	2.1119369	4.1225012	4.7704742
## 75	3.6312198	3.7383711	3.0536972	4.1906073	3.5919719	3.4828027	4.2564684
## 76	3.3340937	3.8277937	3.1932005	3.8664234	3.7391283	3.2488436	4.4742315
## 77	3.4230595	3.0559775	2.8628771	2.5776763	1.7119427	3.6204144	4.3845445
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## 34 4.1778245 4.6492518 5.4942507 4.5505392 3.0328078
## 35 5.0645283 4.5273216 5.6403109 4.9478623 3.4282684 3.4605513
## 36 4.9952087 2.0006876 3.7287671 1.9800494 4.0195449 5.5613107 4.8927668
## 37 4.1615060 2.8880263 3.2736865 2.6186234 3.5832774 4.4495694 4.6008728
## 38 5.7383958 3.0120166 3.0154689 3.2382959 4.5996163 6.3186120 6.6994194
## 39 5.4579496 4.3167193 5.2646645 4.2667194 3.8907346 5.1264921 5.3510564
## 40 4.5373009 5.3269304 6.1808013 5.2735013 4.8819527 5.2293087 5.4757127
## 41 6.1419780 4.7599183 5.5404850 4.0966513 4.1384506 6.1191537 6.5943034
## 42 4.3539791 3.7569705 4.1598585 3.9906302 2.2738904 3.5280751 2.7772436
## 43 4.5093524 1.5930204 2.5512488 2.1327248 3.3536055 5.0038973 4.8625051
## 44 5.5785945 4.5890620 4.2619615 5.2396259 2.6932886 4.4377330 4.3614892
## 45 4.8044884 5.0503345 5.7277350 5.3979346 4.0643306 5.5884087 4.0167177
## 46 4.6268068 5.0078919 5.8316599 5.1296215 4.0919713 5.5700547 4.1456639
## 47 3.5655797 5.4811751 6.3836983 5.6182900 5.4362342 5.8209346 5.1461903
## 48 4.5991218 3.0347082 3.5064181 2.6744717 2.9567531 4.5174421 4.8710459
## 49 4.3191649 1.5660009 3.2246045 1.6010867 3.1364189 3.7280160 3.9291232
## 50 3.3290884 4.9338632 6.1791012 4.5502091 3.9259338 4.2184418 4.1253894
## 51 4.5572681 4.7559481 5.1632533 4.6256549 1.7357454 3.4291001 4.8671542
## 52 3.1129519 3.6650105 4.9141799 3.8005158 3.6610158 3.9451448 3.0696264
## 53 1.8070345 5.4139382 5.8425573 5.4930081 4.6352765 5.1981670 5.0866764
## 54 5.8219291 5.6599949 6.5012597 4.8996197 4.5391911 5.2492191 6.5395717
## 55 8.8612846 6.4041810 6.4514220 6.9403249 5.5520358 6.6464228 7.1811777
## 56 8.5803969 6.5955922 6.4235403 7.1595515 5.3195999 6.5071047 6.9405587
## 57 3.8684833 4.1083059 4.6413675 4.3395019 1.9905402 2.1955921 2.7251609
## 58 5.6965551 5.1856845 4.6406142 5.8869201 3.9156246 4.8305276 4.0156472
## 59 1.7915195 4.8876979 5.2107375 4.8098926 4.1812777 4.8577454 4.9478069
## 60 3.8799020 3.6025994 4.5147084 3.9523030 2.6304848 3.3358661 2.2361839
## 61 4.7604256 4.0265056 4.2537534 4.8703339 2.8199668 2.8589834 3.8153319
## 62 6.2013520 4.4251364 5.1383800 3.7263417 4.5149555 5.4343651 6.5930963

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## 63	5.6524944	4.2791033	4.9859445	3.3212810	4.1906154	4.8726519	6.2167172
## 64	6.8937719	5.5598314	5.0409545	6.0767830	4.1183945	5.2602604	5.9429641
## 65	6.1673269	6.1051340	5.6810768	6.5104494	4.1957833	4.3746795	5.3961794
## 66	6.2640668	5.9380844	5.5475169	6.3405693	4.1578763	4.3149765	5.3931149
## 67	4.7606252	1.8021502	1.9020758	3.4441264	3.8144838	4.9867856	4.4423329
## 68	5.6797173	5.8447812	5.4645335	5.5966592	4.1380172	5.0203029	6.0150504
## 69	5.0802362	4.0459525	3.6736454	4.8248515	2.5082263	4.2183802	4.8583341
## 70	6.1101816	4.9439514	6.0243309	4.5900423	4.3283836	5.2865065	5.8166122
## 71	4.3033964	6.5606448	7.0004054	6.5656693	6.2250434	7.0793556	6.9434242
## 72	5.2052311	5.2711936	5.9223220	5.0449992	3.5499864	4.8384886	5.2399777
## 73	4.7412021	3.6828587	5.2649189	2.8160941	2.8807276	3.3484710	4.3709631
## 74	5.0269574	1.0760994	2.4800270	2.2013586	3.6732606	5.2354511	5.0662364
## 75	4.4466826	4.1773832	4.6179710	3.7021774	2.9030809	3.3144766	4.1914312
## 76	4.7259600	4.2469746	4.4111371	3.9558755	2.6752277	4.1994323	4.7529516
## 77	4.5339702	2.2557866	3.0811520	2.0752825	3.2168858	4.0139254	4.3870293
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## 37	2.8765683						
## 38	2.9317075	3.5708528					
## 39	4.3894047	4.8138149	4.8391127				

## 40	5.4595516	5.4166817	6.1877533	3.1363226			
## 41	4.1710235	4.7444018	3.4506518	4.5070765	5.8964767		
## 42	4.0092559	2.6476697	5.0760018	4.6742387	5.2931792	5.1380876	
## 43	1.6376969	2.2586585	2.3867817	4.1175339	5.2247795	4.0028817	3.3789110
## 44	5.2045062	4.4483990	4.9670342	4.8060371	5.6904896	4.6212855	2.9834307
## 45	4.5514393	4.7587873	5.9497269	5.3074240	4.9860091	5.6197679	3.9396288
## 46	4.3325890	4.5289868	5.8557478	5.2394823	4.9134638	5.4349869	3.9223197
## 47	5.3308686	5.4610095	6.6705302	6.1922452	4.2351904	6.8300862	5.4077043
## 48	2.9183940	2.1204018	2.7741835	4.2838715	5.4570980	3.2620068	3.0357944
## 49	2.2976057	2.2264991	3.4460560	4.0327393	4.6857101	4.2910334	3.0351038
## 50	4.7823533	4.6740276	6.0648942	5.0555036	3.6833270	5.2450533	4.2385890
## 51	5.1457994	4.5383039	4.9578115	4.3191093	5.4522457	3.8663091	3.6071555
## 52	3.8157209	3.6181151	5.6198105	4.9147663	4.1227884	5.9271916	3.2075967
## 53	5.3340475	4.6863312	6.4484140	6.1464709	5.2696237	7.0699855	4.6551598
## 54	5.6618981	5.3423813	5.6774842	2.5557246	4.0319800	4.4942206	5.3952083
## 55	7.2086826	7.3061339	6.7754140	7.2818784	9.5799551	6.8171572	6.4958336
## 56	7.3180276	7.0988357	6.9736255	7.3746219	9.5813652	7.0714776	6.0964429
## 57	4.8110712	3.6076632	5.7372270	4.6696366	5.0392056	5.6314197	1.8429332
## 58	5.7348924	4.2706684	6.1032293	6.5739967	6.7782288	6.3578220	2.7357878
## 59	4.7794843	3.8111558	5.5584717	5.6624971	4.7827456	6.0717919	4.0752897
## 60	3.9827259	3.3535986	5.6169060	4.7838891	5.3184343	5.9876659	1.9517703
## 61	5.3613057	4.8437534	5.6417905	4.9789273	5.7358942	6.1207428	3.7254037
## 62	4.5399590	4.5729053	3.4356240	4.8599920	5.9994994	2.4953254	5.3605097
## 63	4.3786609	3.8195311	3.5923234	4.7537963	5.6667468	2.6934420	4.7401818
## 64	6.2792271	5.5495605	5.4082679	6.5248391	7.9222672	5.4927463	4.8294158
## 65	6.9549948	5.6996867	6.4845031	6.8686310	7.3736474	6.3109790	4.6395858
## 66	6.8488231	5.6384262	6.3244921	6.7395301	7.2702650	6.0884289	4.6069224
## 67	2.8866125	3.0307287	3.7302890	4.9032188	5.6735058	5.7612782	3.5078278
## 68	5.9953973	4.0502538	5.4905532	5.8166314	6.1988622	4.8356143	3.7772026
## 69	4.8868010	4.4628268	4.2599641	4.6928751	5.9166831	4.6853923	3.7394257
## 70	5.0266531	5.4121522	5.2990631	1.4136048	3.4509444	4.2244005	5.2134489
## 71	6.2635191	6.2422854	6.9306240	5.0831610	3.2788353	7.2056696	6.4733233
## 72	5.2233580	4.9453563	5.6319770	1.8735755	3.6310611	4.8450408	4.3206870
## 73	3.9110377	4.0118090	4.6701071	3.8833515	4.7244894	3.4534707	3.8180252
## 74	1.6476300	3.0567080	2.1939372	4.1782303	5.4320990	4.0918923	4.0047286
## 75	4.4222190	2.7085952	4.7032990	4.8330250	5.4329284	4.4456524	2.5825381
## 76	4.2397232	2.9972376	4.0038362	4.5949460	5.5129166	3.4789444	2.8613345
## 77	2.7132120	1.7630684	3.0688015	4.3127679	5.0884346	3.9532531	2.9378633
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## 46 4.3344152 4.8460318 0.6898925
## 47 5.3547858 6.3136842 4.0126595 3.9381784
## 48 2.5095492 3.7443559 4.9799191 4.8383094 5.9461282
## 49 1.8600042 4.1572561 4.4743687 4.3347306 4.9422257 2.6188251
## 50 4.8079663 5.1521682 3.7454854 3.5575436 2.5930832 4.6452932 3.9630024
## 51 4.3969271 3.1087285 5.3330108 5.3316715 6.4653064 3.4300387 4.1794462
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## 53 5.0359210 6.1105354 4.5762252 4.4492365 3.6356900 5.2733314 5.1148439
## 54 5.2360026 5.6357479 6.5770003 6.3516794 7.1988698 4.7983401 4.9421064
## 55 6.5973156 5.6028699 8.5259066 8.6816737 10.5078824 6.2893545 6.7160527
## 56 6.5985290 5.3331787 8.2463444 8.4113558 10.3968316 6.2650006 6.8162088
## 57 4.0094710 3.2075929 4.3199945 4.3460488 5.3534665 3.9179528 3.3170928
## 58 4.9662115 2.9889959 4.6294042 4.8795680 6.2505766 4.4121517 4.7033928
## 59 4.4427439 5.3534921 4.3013727 4.1523787 3.4112366 4.1862617 4.4143649
## 60 3.5798534 4.0842113 4.0072798 3.9884668 5.0764777 3.9035229 3.2030939
## 61 4.5481325 3.4555657 5.6207124 5.8402097 6.2600067 4.4477835 3.9921624
## 62 4.3260462 4.8710611 6.5558507 6.4290968 7.0925038 3.0024840 3.9226969
## 63 4.0151122 4.7081928 6.2296053 6.0279170 6.7287973 2.6471034 3.4438823
## 64 5.5907273 3.5096014 6.8683510 7.0768520 8.4180449 4.1971939 5.5060947
## 65 6.2382642 3.6125896 6.6286790 6.8583117 7.5802748 4.7335606 5.6843285
## 66 6.1043823 3.4168645 6.6288004 6.8547270 7.5585071 4.6450510 5.4946072
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## 68 5.0184863 3.7523676 5.7378417 5.6366323 6.9824061 4.0553269 4.8168686
## 69 4.0417102 2.5671968 5.4640982 5.6797602 6.6119313 3.2835273 4.1311404
## 70 4.8218628 5.0702371 5.8951317 5.7997239 6.6796782 4.6436471 4.4384134

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## 71	6.0791315	7.1052080	5.5202590	5.4039548	4.1317876	6.4864897	6.2668337
## 72	4.8405919	4.6052297	5.3659111	5.2834405	6.5091534	4.3530763	4.7551533
## 73	3.7056482	4.2731639	5.0224231	4.8305742	5.6020455	3.3282532	2.6718299
## 74	1.1489100	4.4917189	4.9434636	4.8912574	5.6502071	2.8234277	2.0678316
## 75	3.9837746	3.6984686	5.1604340	5.0375885	5.9849214	2.0930862	3.2365551
## 76	3.7255795	3.1176811	4.8950727	4.8200974	6.1437989	1.5999616	3.5358723
## 77	2.2793388	3.9249811	4.8666962	4.7421921	5.4799890	1.4806719	1.5541927
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## 55 9.0772652 5.3463153 8.4618025 9.4161325 7.8444334
## 56 9.0480004 5.1967078 8.3061242 9.0152868 7.9072296 1.3736816
## 57 4.0453860 3.0922490 3.1456211 4.4697133 5.1784959 6.3060827 5.9347312
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## 59 3.3778121 4.9807651 3.1807856 1.6288040 6.3106067 9.0930315 8.7806230
## 60 4.2317722 4.1435342 2.6000838 3.8692923 5.8340290 6.6293828 6.2682322
## 61 5.2331208 3.3702710 4.4796612 5.3570490 5.9756109 5.1815327 5.0897392
## 62 5.4269741 4.0912297 6.0842920 7.3383066 4.7683329 6.6882552 7.1282589
## 63 4.9807401 3.8573477 5.5523293 6.8638881 4.3280522 6.9533642 7.2333981
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## 72 5.1103134 3.7285983 5.1849191 5.8167239 2.4107712 7.2943948 7.1302589
## 73 3.5644515 3.0819074 3.9426277 5.8345358 3.9398768 6.4157934 6.6639130
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## 75 4.2626223 3.2071389 4.3167265 5.2161538 4.8840544 6.6569023 6.4962408
## 76 4.5213468 2.7713472 4.7663691 5.4434827 4.7715669 6.3310295 6.1644188
## 77 4.2990289 3.9718049 3.7777202 5.2968316 4.9900388 6.6218119 6.6945606
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## 65 4.4547715 3.8471150 5.9208301 5.4090849 3.7553726 5.4484881 5.3961583
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## 69 2.4613690 3.1314258 3.1070903 4.0660743 4.8842238
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## 71 9.1874728 8.6793181 8.7275001 6.4573390 7.5242320 7.0590904 5.8538432
## 72 6.2071330 6.3334602 6.3070299 5.5817296 5.2411847 4.5655862 2.4116336
## 73 5.4752005 5.4892331 5.2474342 4.8779207 4.9148680 4.3527953 3.5927509
## 74 5.4831321 6.2840756 6.1215339 2.1229868 5.7398169 3.9340692 4.8197413
## 75 4.1582682 3.7635871 3.7137807 4.6662581 3.5132729 3.6313943 4.9854871
## 76 3.6176818 3.7365448 3.6849016 4.6666628 3.2088302 3.0435908 4.7720508
## 77 4.7198991 4.9249124 4.7565723 3.1459072 4.2895232 3.6962567 4.6384473
##      71      72      73      74      75      76
## 2
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## 10
## 11
## 12
## 13
## 14
## 15
## 16
## 17
## 18
## 19
## 20
## 21
## 22
## 23
## 24
## 25
## 26
## 27
## 28
## 29
## 30
## 31
## 32

```

```

## 33
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## 57
## 58
## 59
## 60
## 61
## 62
## 63
## 64
## 65
## 66
## 67
## 68
## 69
## 70
## 71
## 72  5.1520121
## 73  6.7324809  4.2339322
## 74  6.4664707  5.1569687  3.8194767
## 75  6.8106721  4.3931235  3.3007427  4.3899927
## 76  6.6939155  4.1508074  3.5118040  4.1060031  1.4698887
## 77  6.5026113  4.7318239  3.0539597  2.4852793  2.2359726  2.4625095

```

Hierarchical Clustering

```

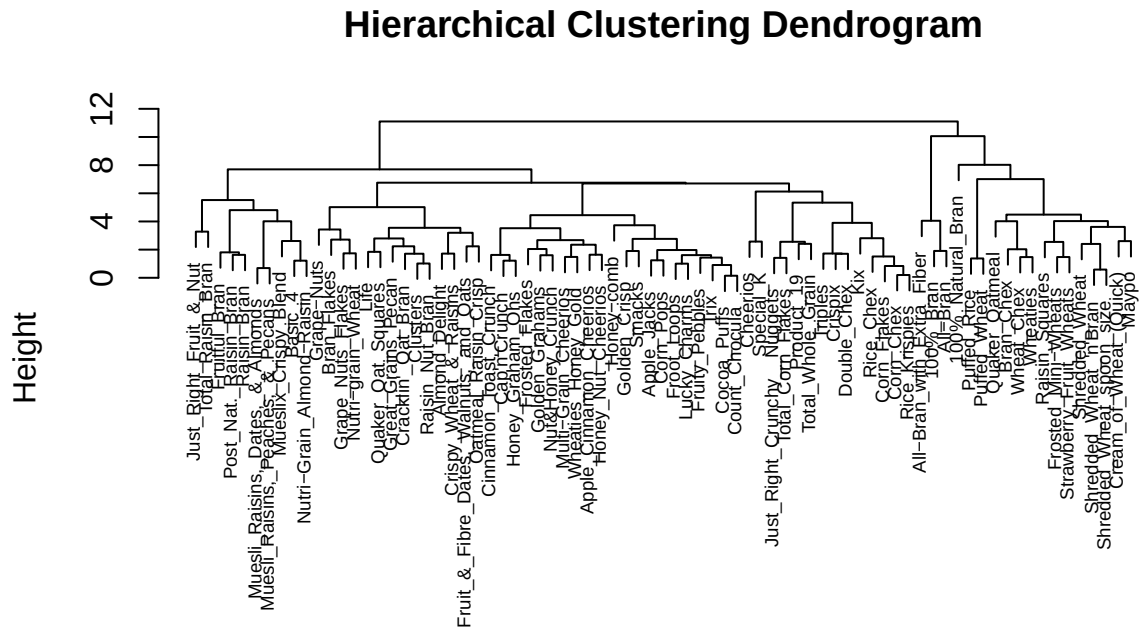
hc_complete <- hclust(
  distance_matrix,
  method = "complete"
)

```

```
)
hc_complete
```

Dendrogram

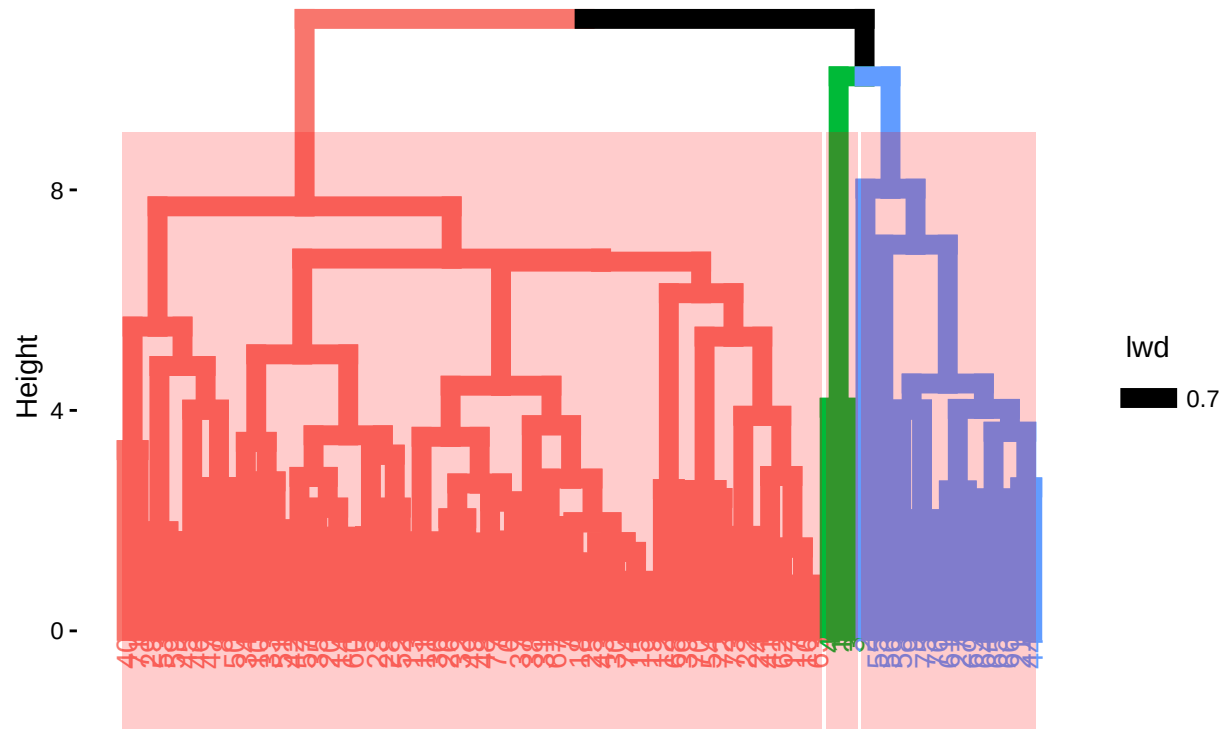
```
plot(
  hc_complete,
  labels = cereals$name,
  main = "Hierarchical Clustering Dendrogram",
  xlab = "",
  sub = "",
  cex = 0.6
)
```



Visualize Cluster Structure

```
fviz_dend(
  hc_complete,
  k = 3,
  rect = TRUE,
  rect_fill = TRUE,
  rect_border = "red"
)
```

Cluster Dendrogram



Assign Clusters

```
cereals$Cluster <- as.factor(
  cutree(hc_complete, k = 3)
)

table(cereals$Cluster)
```

```
##
##  1  2  3
##  3 15 59
```

Cluster Profiles

```
cluster_summary <- aggregate(
  cereals_numeric,
  by = list(Cluster = cereals$Cluster),
  FUN = mean
)
```

```
cluster_summary
```

```
##   Cluster  calories  protein      fat   sodium    fiber    carbo  sugars
## 1      1    63.33333 4.000000 0.6666667 176.66667 11.000000  6.666667 3.666667
## 2      2    90.00000 2.733333 0.7333333  49.33333  2.313333 15.366667 3.200000
## 3      3   113.38983 2.423729 1.1016949 186.86441  1.661017 15.067797 8.169492
##      potass vitamins  shelf  weight    cups  rating
## 1 310.00000 25.00000 3.000000 1.000000 0.3866667 73.84446
## 2 100.00000 11.66667 1.800000 0.922000 0.8433333 57.80076
## 3  87.28814 32.62712 2.271186 1.058475 0.8374576 37.23245
```

Additional Cluster Interpretation

```
table(cereals$Cluster, cereals$mfr)
```

```
##
##      A  G  K  N  P  Q  R
## 1  0  0  2  1  0  0  0
## 2  1  1  2  5  0  4  2
## 3  0 21 19  0  9  4  6
```

```
table(cereals$Cluster, cereals$type)
```

```
##
##      C  H
## 1  3  0
## 2 12  3
## 3 59  0
```

```
table(cereals$Cluster, cereals$shelf)
```

```
##
##      1  2  3
## 1  0  0  3
## 2  7  4  4
## 3 13 17 29
```

Interpretation Using Non-Clustering Variables

Cereal Type

Cluster 1 contains only cold cereals. Cluster 2 contains both cold and hot cereals. Cluster 3 contains only cold cereals. Hot cereals appear concentrated within Cluster 2, suggesting they share similar nutritional

characteristics.

Shelf Placement

Cluster 1 cereals are found exclusively on Shelf 3. Cluster 2 cereals are distributed across all shelves. Cluster 3 cereals dominate Shelf 3 and represent the majority of products throughout the store.

This suggests that shelf placement may reflect product popularity and market positioning.

Cluster Interpretation

Cluster 1 (3 Cereals)

Cluster 1 contains only three cereals and represents the healthiest cereals in the dataset. These cereals have the lowest calorie content, the highest protein content, and exceptionally high fiber and potassium levels. Their relatively low sugar content suggests that these cereals are designed for health-conscious consumers seeking high nutritional value and dietary benefits.

Cluster 2 (15 Cereals)

Cluster 2 represents moderately healthy cereals. These cereals have lower calories and sodium levels than Cluster 3 while maintaining moderate fiber and carbohydrate content. They also contain the lowest average sugar content among the clusters. This group appears to target consumers seeking a balance between taste and nutritional quality.“

Cluster 3 (59 Cereals)

Cluster 3 is the largest cluster and contains the majority of cereals in the dataset. These cereals have the highest calorie levels, highest sodium levels, and highest sugar content. Fiber and protein levels are relatively low compared to Cluster 1. This cluster likely represents mainstream commercial cereals designed primarily for taste and mass-market appeal rather than nutritional optimization.

Findings

The hierarchical clustering analysis identified three distinct groups of cereals based on nutritional characteristics. Cluster 1 consisted of a small group of highly nutritious cereals characterized by high fiber, protein, and potassium content with low calories and sugars. Cluster 2 represented moderately healthy cereals with balanced nutritional profiles and the presence of both hot and cold cereal products. Cluster 3 contained the majority of cereals and was characterized by higher calories, sodium, and sugar levels, reflecting typical mass-market cereal products. The results demonstrate that hierarchical clustering can effectively segment cereal products into meaningful nutritional groups that may assist manufacturers, retailers, and consumers in product evaluation and decision-making.

Conclusion

This assignment applied hierarchical clustering techniques to a cereal nutrition dataset consisting of 77 cereal products. After handling missing values and standardizing the numerical variables, Euclidean distance and complete linkage methods were used to construct a hierarchical clustering model.

The dendrogram and cluster analysis revealed three meaningful clusters that differed substantially in terms of calories, protein, fiber, sugar, sodium, and potassium content. Cluster 1 represented highly nutritious cereals, Cluster 2 represented moderately healthy cereals, and Cluster 3 represented mainstream cereal products with higher calorie and sugar levels.

Overall, the analysis demonstrates that hierarchical clustering is a valuable unsupervised machine learning technique for identifying natural groupings within complex datasets and providing actionable insights into product characteristics and consumer market segments